

MODULE 09 · DOMAIN 1 - AI & ML FUNDAMENTALS

Model Customisation & SageMaker AI

The build-it-yourself tier, and the tools that make it survivable.

DOMAIN 1 of 5 EXAM WEIGHT 20% MODULE 9 of 14

Bedrock gives you a model. SageMaker gives you **a workshop**. The exam asks which one the scenario needs.

Bedrock or SageMaker?

BEDROCK

Use a foundation model

- Generative tasks: text, images, chat, code
- No training data required to start
- Serverless, priced per token
- Customise by prompt, RAG or fine-tuning
- No ML expertise needed

SAGEMAKER AI

Build your own model

- Traditional ML: classification, regression, forecasting
- You have labelled training data
- You choose the algorithm and the instances
- Full control of training and deployment
- Needs data science capability

THE TIEBREAKER

Generative task means Bedrock. Predicting a number or category from your own tabular data means SageMaker.

SageMaker across the ML lifecycle

STAGE	THE TOOL
Label data	Ground Truth
Prepare and explore data	Data Wrangler, Canvas
Store reusable features	Feature Store
Start from a pre-trained model	JumpStart
Build without code	Canvas
Train and tune	Training jobs, Automatic Model Tuning
Deploy	Endpoints: real-time, serverless, async, batch
Watch in production	Model Monitor
Explain and check bias	Clarify
Document and govern	Model Cards, Model Registry

ON THE EXAM

Most SageMaker questions are one row of this table. Identify the lifecycle stage in the scenario and the tool follows.

JumpStart and Canvas

- ◆ **JumpStart** — a catalogue of pre-trained models and solution templates you can deploy or fine-tune.
- ◆ Covers open-source foundation models as well as traditional ML models.
- ◆ **Canvas** — a visual, no-code interface for building models from tabular data.
- ◆ Canvas is the answer for **business analysts with no coding skills**.
- ◆ Canvas now also covers data preparation, absorbing Data Wrangler's capabilities.

ON THE EXAM

'Business analyst', 'no coding experience' or 'without writing code' points at Canvas. It appears often and is rarely a distractor.

Ground Truth, Data Wrangler and Feature Store

- ◆ **Ground Truth** — creates labelled training data using human labellers.
- ◆ Workforces can be your own staff, a vendor, or Mechanical Turk.
- ◆ **Data Wrangler** — import, explore, clean and transform data with little code.
- ◆ **Feature Store** — a central repository of engineered features, reused across models.
- ◆ Feature Store prevents **training-serving skew**: features computed the same way in both places.

ON THE EXAM

'We need labelled data' means Ground Truth. 'The same features used by several teams and models' means Feature Store.

Training jobs and tuning

- ◆ A **training job** runs your algorithm on your data using instances you specify.
- ◆ **Hyperparameters** are settings you choose before training: learning rate, epochs, batch size.
- ◆ An **epoch** is one complete pass through the training dataset.
- ◆ **Automatic Model Tuning** searches hyperparameter combinations to find the best result.
- ◆ Hyperparameters are set by you. Parameters are learned by the model. The exam tests this distinction.

ON THE EXAM

Asked how to improve a model without changing the algorithm or the data, the answer is hyperparameter tuning.

Four ways to serve a model

OPTION	USE IT WHEN
Real-time endpoint	Low latency, steady traffic. Always on, always billing.
Serverless inference	Intermittent or unpredictable traffic. Scales to zero; cold starts.
Asynchronous inference	Large payloads or long processing. Requests are queued.
Batch transform	A whole dataset at once. No endpoint at all.

ON THE EXAM

Intermittent traffic with idle periods means serverless. A whole dataset with nobody waiting means batch transform.

Model Monitor and drift

- ◆ Monitors a deployed model for **drift** and quality degradation over time.
- ◆ **Data drift** — the input distribution changes. Your customers are not who they were.
- ◆ **Concept drift** — the relationship between input and outcome changes.
- ◆ Detects bias appearing after deployment, not just at training time.
- ◆ Alerts through CloudWatch, so retraining can be triggered.

ON THE EXAM

Any scenario about a model getting worse months after deployment is a Model Monitor question. Drift is the mechanism.

Clarify does two jobs

- ◆ **Bias detection** — checks datasets and models for imbalance across groups.
- ◆ Runs before training on the data, and after training on the predictions.
- ◆ **Explainability** — feature importance and SHAP values explaining individual predictions.
- ◆ Produces the **partial dependence plots** used in transparency reports.
- ◆ Bridges Domain 1 and Domain 4, appearing in both.

ON THE EXAM

Clarify is the single answer to both 'is this model fair' and 'why did it decide that'. Remember it does both jobs.

Model Cards, Registry and Pipelines

- ◆ **Model Cards** — documented intended use, training data, metrics, limitations and risk rating.
- ◆ **Model Registry** — versioned models with approval status before deployment.
- ◆ **Pipelines** — automate the whole flow: prepare, train, evaluate, register, deploy.
- ◆ Together these are what **MLOps** means on AWS.
- ◆ Model Cards document a model you built. **AI Service Cards** document AWS's own services.

ON THE EXAM

That last distinction is tested directly. Model Card is yours; AI Service Card is Amazon's, published for services like Rekognition.

Studio, Unified Studio and Catalog

- ◆ **SageMaker Studio** — the integrated development environment for ML work.
- ◆ **SageMaker Unified Studio** — one workspace spanning data, analytics and AI tools.
- ◆ **SageMaker Catalog** — discover and govern data and model assets with permissions.
- ◆ Catalog is the governance answer: who may use which dataset or model.

Ten SageMaker features closed on 30 July 2026

- ◆ **Ground Truth, Augmented AI, Model Monitor, Clarify** — all closed to new customers.
- ◆ Also Debugger, Profiler, Role Manager, Studio Lab, Geospatial and Mechanical Turk integration.
- ◆ **Ground Truth Plus** ended entirely on 30 June 2026.
- ◆ Core SageMaker is **not** affected: Training, Inference, Studio and Canvas continue.
- ◆ All ten remain correct answers on the current exam.

ON THE EXAM

Clarify and A2I are taught as correct answers earlier in this course, and they still are. Availability changed; the exam did not.

Carry these into the exam

- ◆ Generative task means Bedrock. Your own tabular data means SageMaker.
- ◆ 'No coding experience' or 'business analyst' means Canvas.
- ◆ Hyperparameters are set by you; parameters are learned by the model.
- ◆ Intermittent traffic means serverless inference. Whole dataset means batch transform.
- ◆ Model degrading after months in production means Model Monitor and drift.
- ◆ Clarify does bias AND explainability. Model Card is yours; AI Service Card is Amazon's.

A retailer wants to predict which customers will churn next month, using five years of its own transaction and support history.

The analytics team knows SQL and Excel but has no machine learning engineers, and does not want to write code.

What should they use?

- ☐ A Amazon Bedrock with a foundation model
- ☐ B SageMaker Canvas
- ☐ C SageMaker training jobs with a custom algorithm
- ☐ D Amazon Q Business

◆ PAUSE AND DECIDE

What should they use?

- A

 Bedrock
Generative models. This is prediction from tabular data.
- B

 SageMaker Canvas
No-code model building on their own tabular data.
- C

 Custom training jobs
Correct tier, but needs the engineers they do not have.
- D

 Q Business
Answers questions from documents. Not prediction.

ON THE EXAM

Two signals: tabular prediction rules out Bedrock, and 'no code' rules out custom training. Canvas sits exactly between them.

A credit model deployed eighteen months ago passed all its tests at launch. Approval rates have slowly shifted and the risk team reports the model no longer matches recent outcomes.

Nothing about the model or its code has changed.

What is happening, and which service addresses it?

- | | |
|---|---|
| ☐ A Overfitting; retrain with regularisation | ☐ B Drift; SageMaker Model Monitor |
| ☐ C Hallucination; add guardrails | ☐ D Underfitting; use a larger model |

◆ PAUSE AND DECIDE

What is happening, and which service addresses it?

- A

 Overfitting
Would have shown at launch. It passed its tests.
- C

 Hallucination
A generative model failure. This is a credit classifier.
- B

 Drift; Model Monitor
The world changed while the model stayed still.
- D

 Underfitting
Would also have shown at launch, not eighteen months later.

ON THE EXAM

'Passed its tests at launch' plus 'nothing changed' equals drift. Both overfitting and underfitting are visible on day one.

A regulator asks a lender to demonstrate two things: that its model does not disadvantage applicants by age, and that any individual rejection can be explained.

The model runs on SageMaker.

Which service produces both?

- ☐ A SageMaker Model Monitor
- ☐ B SageMaker Clarify
- ☐ C SageMaker Model Cards
- ☐ D Two services are needed

◆ PAUSE AND DECIDE

Which service produces both?

- A

 Model Monitor
Watches production drift. Does not explain a decision.
- B

 Clarify
Bias detection and per-prediction explanation in one service.
- C

 Model Cards
Documents the model. Does not analyse it.
- D

 Two services needed
Tempting, but Clarify covers both requirements.

ON THE EXAM

Option D is the trap for anyone who remembers Clarify does only one of the two jobs. It does both.

A pathology startup deploys an image classification model. Requests arrive a few times an hour during clinic sessions and not at all overnight.

A real-time endpoint currently runs continuously, and the bill is dominated by idle time.

Which deployment option fits best?

- | | |
|--|---|
| ☐ A Serverless inference | ☐ B Batch transform |
| ☐ C A larger real-time endpoint | ☐ D Asynchronous inference |

◆ PAUSE AND DECIDE

Which deployment option fits best?

- A

 Serverless inference
Scales to zero between requests. Idle time stops costing.
- B

 Batch transform
No endpoint, but clinicians need an answer now.
- C

 Larger endpoint
More idle capacity to pay for. Worse, not better.
- D

 Asynchronous inference
For large payloads and long jobs. The problem is idleness.

ON THE EXAM

Intermittent traffic with real idle gaps is the serverless signature. Batch is wrong because someone is waiting.